

Structured Securities Proposal and Risk Mitigation Report

Selected Scenario: Conservative Investor

Client Objective:

Seeks stable returns with limited risk exposure, while gaining modest exposure to coffee price movements. The investor prioritizes capital preservation and predictable income over high upside potential.

Recommended Product: Commodity-Linked Bond

Quantitative Model Used

Pricing Model:

$$P_b = P_f + \lambda \bullet \Delta C$$

Where:

- P_b = Bond return
- P_f = Fixed interest component
- λ = Sensitivity to coffee price changes
- ΔC = Change in coffee price ($C_T - C_0$)

Best Practice and Justification

Quantitative Justification

This structure provides a hybrid payoff combining:

1. A deterministic fixed income component P_f , ensuring income stability.
2. A linear commodity-linked component $\lambda(C_T - C_0)$ allowing controlled participation in coffee price movements.

This structure is appropriate for a conservative investor because it ensures:

- A guaranteed fixed interest component
- Limited exposure to commodity price volatility
- A bond-like risk profile

The fixed component P_f provides income stability and capital preservation, which aligns with the client's low-risk preference. The sensitivity parameter λ allows controlled participation in coffee price movements without exposing the investor to direct commodity trading risks.

By carefully calibrating λ , we can limit volatility while still offering incremental returns if coffee prices rise.

Quantitative Implementation

To determine the appropriate bond structure:

1. Estimate expected coffee price movement.
2. Adjust λ to control commodity sensitivity.
3. Set P_f at a level that ensures stable income while maintaining overall bond attractiveness.

Risk analysis can be performed by evaluating various coffee price scenarios to ensure downside exposure remains within the investor's tolerance.

Key Benefits

- Stable and predictable income
- Lower volatility compared to structured notes or digital options
- Simpler pricing structure
- Reduced downside risk
- Suitable for capital preservation strategies